BIOT6900 Assignment 6 Review

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Background

In this assignment, I calculated Polygenic Risk Scores (PRS) using 1000 Genomes genotypes (1000 Genomes Project Consortium, 2015) and a published Alzheimer's scoring system from the PGS Catalog (PGS Catalog, n.d.; Zhang et al., 2020). Along with the provided data, I harmonized the genotype data with known Alzheimer's variants, identifying which variants were represented as reference or alternate allele in the genotype data. Then, calculating the dosages per allele, I created a PRS. Using the PRS along with a simulated phenotype, I stratified by ancestry to judge the accuracy of the PRS.

Accuracy loss in other ancestries

The dataset skews heavily toward European ancestries. The Alzheimer's score from the PGS Catalog was developed from a GWAS study that skewed at least 80.9% European (the remaining 19.1% were not reported; PGS Catalog, n.d.). This means that the scores associated with the variants (dosage × log(OR)) are biased towards the effects on those with European ancestry, and these effects may not replicate across different ancestry groups. One gene which is highly correlated with Alzheimer’s in those of European ancestry may have an even higher correlation in another background, or no correlation at all. Because of this, I observed a drop in accuracy when stratifying by ancestry. The PRS scored an AUC of 0.672 on those with European ancestries, its highest ancestry-specific score. Its accuracy declined in other groups though, with the AUC for East Asians being 0.575, 0.611 for South Asians, and 0.555 for Admixed Americans. It scored the worst against the African population group, actually performing slightly worse than random chance, with an AUC of 0.493.

This effect can be mostly explained by Linkage Disequilibrium (LD) and the differences in allele frequency among groups. Linkage Disequilibrium is the genetic observation that nearby variants are inherited together. This creates an issue for developing PRSs, as counting each individual associated variant by itself would multiply the signal in highly correlated areas, leading to the model scoring individual variants as having much higher risk than they do in reality. To account for this, PRSs are designed by clumping together nearby variants (PGS

Catalog, n.d.) . The connection to ancestry arises in the fact that the minutiae of LD effects are not necessarily shared across all ancestries. Variants which are associated in European genetics may not be so in other backgrounds. This leads to a bias in the model which reduces performance on non-European backgrounds. LD interacts with the other major source of error: allele frequency between groups. Certain alleles which are causal in one group may not be as prevalent in other groups, throwing off predictions. Accounting for these effects is a crucial component of designing a PRS.

One threshold, many groups

It is clear from the results that applying one arbitrary 10% cutoff to the whole-cohort is not a sound method. Because of the previously discussed issues with portability between ancestry groups, some ancestry groups may be scored inaccurately high or low. With a blanket 10% cutoff, this nuance is lost and each group is judged together rather than separately. From the example, it is clear why this is an issue: the PRS scores 28% of those in the African ancestry group above the cutoff, which is far too large a number to be realistic (the African ancestry group scored a 0.493 AUC, so this is an especially dubious prediction). Additionally, East Asians are underflagged at only 2.2% above the cutoff. If a clinic applied this broad cutoff without stratifying by ancestry group, they would fallaciously assign higher or lower risk to groups which in reality may or may not face such risk.

Discrimination vs calibration

As previously outlined, the PRS scores drop precipitously for non-European groups. By discriminating by ancestry, I can get an idea of if the score is ranking people correctly (i.e. if people with higher PRS scores actually tend to develop the condition more often than those who have lower scores)

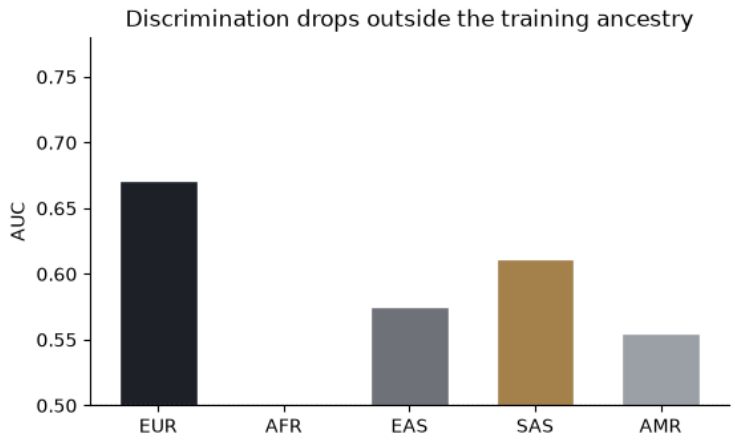


Figure 1: *Discrimination by Ancestry Graph (from the notebook)*

I also calibrated the scores by ancestry: applying the European ancestry derived model to each group, judging it against observed risk within those groups. This gives an idea of how numerically close the predictions actually are (i.e. given a score, does the models predicted probability actually match the observed probability of that group)

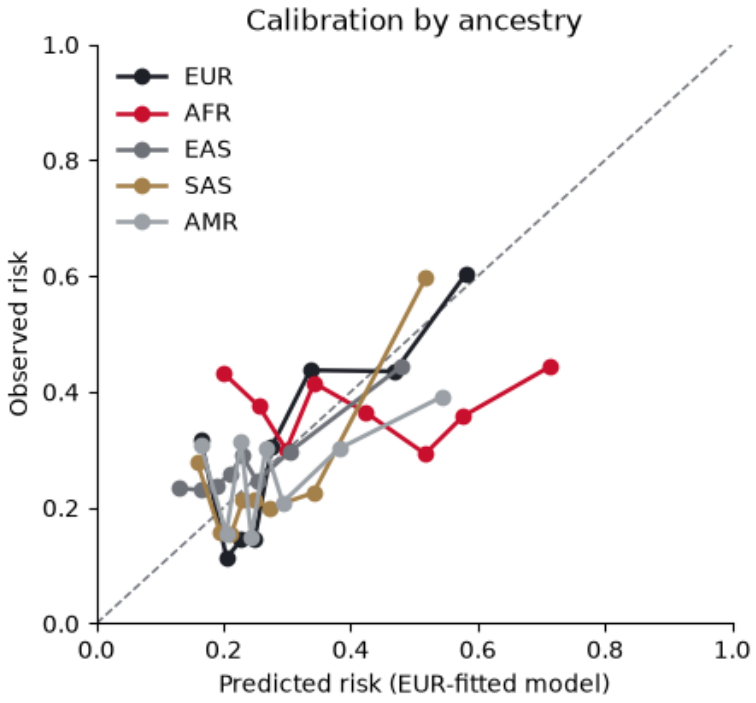


Figure 2: *Calibration by Ancestry Graph (from the notebook)*

Observing the calibration graph, the European data points lie relatively close to the diagonal, indicating that the observed risk matches the predicted risk for this group. For the East Asian and South Asian groups, the calibration gives a slightly more optimistic result for the effectiveness of

the PRS: the data points hover somewhat close to the diagonal. The accuracy is lower for the Admixed American group however, and the calibration is wildly off for the African group (this matches the discrimination metric). Overall, the calibration indicates a better performance for East Asians and South Asians than the discrimination figure, while both predicting the same poor performance for Admixed American and African populations.

Clinical utility and limits

A PRS is not a diagnosis. When applied to individuals, a PRS is simply a (biased) estimation of their likelihood of facing certain health risks, and does not mean that the individual will actually be diagnosed with that disease. Crucially, most diseases are affected by environmental factors which are not present in the PRS data. Instead, the score can be used on an individual level to allow for targeted interventions for individuals at high risk. For this to be a valid use case however, the data cohort needs to be representative of those whom the scores are being applied to, and LD and allele frequency effects need to be accounted for. In this assignment, I used a dataset with simulated phenotypes. To develop a PRS with real clinical utility, data with real phenotypes (preferably also with age, ancestry, and other covariates) would be required.

Proof of real data

source : REAL
people : 2336
score size : 22 variants
by ancestry : {'AFR': np.int64(500), 'EAS': np.int64(500), 'EUR': np.int64(500), 'SAS': np.int64(489), 'AMR': np.int64(347)}
genotypes.csv: sha256 9cf41a1f
variants.csv: sha256 f7a87d3f
score.csv: sha256 7f80e8ff
meta.csv: sha256 6cc55b6e

References

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1000 Genomes Project Consortium. (2015). A global reference for human genetic variation. *Nature, 526*(7571), 68–74. https://doi.org/10.1038/nature15393

PGS Catalog. (n.d.). *PGS000334 (GRSfull_22)* [Polygenic score]. Retrieved October 30, 2026, from https://www.pgscatalog.org/score/PGS000334/

Zhang, Q., Sidorenko, J., Couvy-Duchesne, B., Marioni, R. E., Wright, M. J., Goate, A. M., Marcora, E., Huang, K.-L., Porter, T., Laws, S. M., Australian Imaging Biomarkers and Lifestyle (AIBL) Study, Sachdev, P. S., Mather, K. A., Armstrong, N. J., Thalamuthu, A., Brodaty, H., Yengo, L., Yang, J., Wray, N. R., . . . Visscher, P. M. (2020). Risk prediction of late-onset Alzheimer's disease implies an oligogenic architecture. *Nature Communications, 11*, Article 4799. https://doi.org/10.1038/s41467-020-18534-1